Attendance & Project Progress Log (July 1 - July 20, 2026)

This log tracks the daily work, collaborative coding sessions, data collection runs, and theoretical studies completed for the **USRP N210 Channel State Information (CSI) Human Activity Recognition (HAR)** project.

- **Period**: July 1, 2026 to July 20, 2026 (Monday to Friday only).
- **Format**: Daily task entries divided into **Work Planned**, **Work Completed**, and **Technical Study & Research** sections. Tuesdays, Wednesdays, and Thursdays feature more detailed tasks to highlight research investigations, hardware debugging, and model design decisions.
- **Collaborators**: Rishiikesh, Amrith, Gowreesh (collaborative team effort).

Daily Logs

Week 1: July 1 – July 3

Wednesday, July 1 (Extra Tasks Day) — Hardware Workspace Setup & Link Alignment

- **Work Planned**:
  - Establish the physical sensing environment, define geometry parameters, and configure network links for the USRP N210.
- **Work Completed**:
  - Aligned the USRP N210, MacBook Pro, and Android phone (hotspot) on the floor using fixed tape marks for geometric consistency across capture sessions.
  - Configured host and USRP network interfaces (Fedora host IP `192.168.10.1/24`, USRP IP `192.168.10.7` on interface `eno1`).
- **Technical Study & Research**:
  - **Link Geometry**: Measured and documented link parameters. Set MacBook-to-USRP distance to 3.5 meters and antenna height to 1.1 meters (matching the torso level of standing subjects) to ensure direct path signal blocking. Locked Channel 11 (2.462 Hz) as the initial operating frequency.

Thursday, July 2 (Extra Tasks Day) — Physical CSI Principles & Phase Noise Study

- **Work Planned**:
  - Conduct a literature review on CSI fundamentals and evaluate if carrier phase can be used for motion classification.
- **Work Completed**:
  - Studied OFDM physical layer details and the representation of subcarriers.
- **Technical Study & Research**:
  - **OFDM Subcarrier Extraction**: Determined that standard 20 MHz 802.11a/g frames have 64 FFT subcarrier slots, with 12 null/guard subcarriers and 52 active subcarriers.
  - **Phase Noise Analysis**: Studied three phase noise sources: CFO (Carrier Frequency Offset), SFO (Sampling Frequency Offset), and PBD (Packet Boundary Detection) timing errors that randomize phase packet-to-packet. Since a SISO (single antenna) system cannot use differential phase subtraction, we officially decided to drop phase and utilize amplitude only ($|H_k| = \overline{Re_{2k} + _{2k}}$).

Friday, July 3 — GNU Radio Companion Verification

- **Work Planned**:
  - Verify GNU Radio companion setup and analyze Bastian Bloessl's `gr-ieee802-11` block layout.
- **Work Completed**:
  - Checked the installation and compatibility of the `gr-ieee802-11` OOT block. Outlined code changes for `extract_csi_impl.cc` to log data directly to CSV from PDU inputs instead of passing vectors downstream.

Week 2: July 6 – July 10

Monday, July 6 — Implementing Custom MAC Filtering

- **Work Planned**:
  - Modify `extract_csi_impl.cc` to implement dynamic MAC filtering and CSV saving.
- **Work Completed**:
  - Edited C++ files `extract_csi_impl.cc` and `extract_csi_impl.h`. Coded custom PDU handling that reads environment variables `CSI_TA`, `CSI_RA`, and `CSI_OUT` at runtime, filtering frames by MAC and appending to CSV directly.

Tuesday, July 7 (Extra Tasks Day) — Module Compilation & Verification Gates Design

- **Work Planned**:

    - Rebuild the modified `gr-ieee802-11` module and establish QA gates for recording sessions.
- **Work Completed**:
    - Recompiled the module (`make -j$(nproc) && sudo make install && sudo ldconfig`). Verified compilation using `strings` to check environment variable detection.
- **Technical Study & Research**:
    - **QA Gates formulation**: Formulated **6 Verification Gates** to prevent corrupted data collection: Gate 1 (MAC check), Gate 2 (channel-frequency mapping), Gate 3 (macOS gateway ping at `-i 0.01` to enforce 100 pkt/s), Gate 4 (Rate Gate for ≥ 50 rows/s), Gate 5 (Wiggle Test to confirm motion sensitivity), and Gate 6 (Gain/Saturation checking via `uhd_fft`).

Wednesday, July 8 (Extra Tasks Day) — ImgFi Paper Study & Image Conversion

- **Work Planned**:
    - Study the ImgFi paper (Zhang & Jiao, 2023) and write image conversion scripts.
- **Work Completed**:
    - Developed `make_images.py` to convert CSI amplitude matrices into 224x224 grayscale recurrence transformation (RT) images.
- **Technical Study & Research**:
    - **RT Image Math**: Analyzed why RT ($RT[i,j] = |x_i - x_j|$) outperforms other transformations (GASF, GADF, MTF, STFT) by preserving continuous amplitude relationships between times i and j without arbitrary thresholds. Verified that treating each of the 52 subcarriers as a separate image increases training data by 52 without losing temporal relationships.

Thursday, July 9 (Extra Tasks Day) — Session Logger Scripting & Activity Definitions

- **Work Planned**:
    - Develop session recording scripts and standardize activity performance protocols.
- **Work Completed**:
    - Programmed `session_logger.py` supporting both discrete repetition logging (saving start/end timestamps on ENTER) and continuous logging (`--continuous`).
- **Technical Study & Research**:
    - **Standard Activity Rules**: Defined exact protocols for the 7 classes: `walk` (continuous traversal), `squat` (2-3s full range), `bend` (2-3s pick-up motion), `drink` (2-3s sip

motion), `nod` (3-4 clear head movements), `noaction` (sitting/standing still), and `empty` (room vacated).

Friday, July 10 — Verification Gate Testing

- **Work Planned**:
  - Run verification gates on the host systems and resolve environment issues.
- **Work Completed**:
  - Tested MacBook gateway pinging and verified Fedora receiver compatibility under rotating MACs. Handled MAC changes by updating environment variables.

Week 3: July 13 – July 17

Monday, July 13 — Preliminary Calibration Captures

- **Work Planned**:
  - Perform test captures and evaluate packet interval statistics.
- **Work Completed**:
  - Saved calibration data (`csi_noaction.csv`) and verified packet rates under ambient noise conditions.
- **Technical Study & Research**:
  - **Packet Jitter**: Analyzed arrival timestamps, identifying a median gap of 12 ms and maximum gap of 464 ms, highlighting the need for linear time-resampling.

Tuesday, July 14 (Extra Tasks Day) — Window Segmentation & Resampling Design

- **Work Planned**:
  - Write window segmentation scripts and analyze resampling frequencies.
- **Work Completed**:
  - Developed `segment_windows.py` to slice CSI amplitude and resample irregular packets onto a uniform 100 Hz grid using linear interpolation.
- **Technical Study & Research**:
  - **Interpolation Gating**: Evaluated the interpolation trade-off. Since the actual average packet rate is ~66 Hz, windowing with a target of 100 Hz requires a high packet count. Implemented a fill check threshold that skips any window with < 50 of expected packets to prevent artificial signal patterns.

Wednesday, July 15 (Extra Tasks Day) — Multi-Subject Captures & QA Script Development

- **Work Planned**:
  - Capture preliminary multi-subject data and automate QA checks.
- **Work Completed**:
  - Captured test files for Amrith and Rishiikesh (`Amrith_action_test_1.csv` , `Rishikesh_action_test.csv` , `csi_data_noaction_15july.csv`). Coded `check_capture.py` to automate QA checks.
- **Technical Study & Research**:
  - **QA Criteria**: Programmed check criteria in `check_capture.py` to automatically detect row rate drops, NaNs, all-zero columns, and plot amplitude-over-time curves (Wiggle Test verification).

Thursday, July 16 (Extra Tasks Day) — Doppler Shift & Sampling Frequency Study

- **Work Planned**:
  - Calculate Doppler shift parameters and check sampling rate sufficiency.
- **Work Completed**:
  - Conducted spectral analysis on `action` vs `noaction` calibration files.
- **Technical Study & Research**:
  - **Doppler Shift math**: Calculated Doppler scaling: $f_d = \frac{2vf_c}{c}$. At 2.462 GHz, Doppler shift is $16.4\ \text{Hz}$ per m/s of radial movement. Found that discriminative signal energy drops to the noise floor above $18\ \text{Hz}$. Checked Nyquist headroom: our worst-case rate (51.5 Hz) gives a Nyquist frequency of 25.8 Hz, and our median (66.4 Hz) gives 33.2 Hz, providing sufficient headroom for body movements (though fast hand gestures could alias).

Friday, July 17 — Formal Pilot Dataset Collection

- **Work Planned**:
  - Capture the official binary pilot dataset for the action vs noaction study.
- **Work Completed**:
  - Collected 8 official pilot CSVs for Rishiikesh and Amrith (walk, stand, mixed, sit) spanning 90 seconds each. Ran `check_capture.py` on all files to confirm they passed QA (51-78 rows/s, 0 NaNs).

Week 4: July 20

Monday, July 20 — Model Development & Comparative Analysis

- **Work Planned**:
  - Process the pilot dataset, train the ResNet-18 ImgFi baseline, and build the custom CNN-BiGRU-Attention model.
- **Work Completed**:
  - Segmented windows with `--rate 50` to avoid dropped windows. Trained ResNet-18 majority-vote baseline (`Action_NoAction_imgfi.ipynb`) and custom CNN-BiGRU-Attention model (`Action_NoAction_CNN_GRU.ipynb`). Ran leave-one-subject-out (LOSO) cross-validation.
- **Technical Study & Research**:
  1. **Effective Rank**: Calculated $Rank_{eff} = \exp(-\sum \tilde{\lambda}_i \ln \tilde{\lambda}_i)$. Found a rank of 3.8 for action and 11.9 for noaction, proving that subcarriers collapse to ~4 independent dimensions during movement due to a single large perturber.
  2. **Stand vs. Sit**: Observed that `stand` was the hardest class (95.1% acc) due to postural sway and breathing, whereas `sit` was 100% due to mechanical bracing.
  3. **Model comparison**: Verified that our 0.57 M parameter CNN-BiGRU-Attention model matches the 11.2 M parameter ResNet-18 majority-vote accuracy (98.35% test accuracy) while using 19.6x fewer parameters and 52x fewer forward passes (0.55 ms inference).
  4. **Baseline comparison**: Checked a single motion energy scalar threshold (average subcarrier standard deviation), which also got 98.35% accuracy, indicating that the binary task is simple and the custom model's real value lies in sequential multi-class actions.

Attendance Summary Table

Date	Day	Hours	Planned Work Summary	Completed Work Summary	Key Parameters / Notes
July 1	Wed	8.0	Setup physical environment & links	Aligned USRP, MacBook, Android phone; configured IP links	3.5 m line-of-sight path, 1.1 m antenna height, Channel 11
July 2	Thu	8.0	Review OFDM & carrier phase noise	Studied FFT sizes, 52 subcarrier output, and CFO/SFO/PBD offsets	Selected amplitude-only $

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July 3	Fri	6.0	Check GNU Radio module setup	Evaluated Bastian Bloessl's `gr-ieee802-11` module installation	Identified code changes for CSV output
July 6	Mon	7.5	Code dynamic MAC filtering in C++	Modified `extract_csi_impl.cc` to read filter variables	Support `CSI_TA`, `CSI_RA`, `CSI_OUT` env vars
July 7	Tue	8.0	Compile OOT module & design QA gates	Recompiled `gr-ieee802-11` module; defined 6 QA gates	Gates include Rate Gate (≥ 50 rows/s) and Wiggle Test
July 8	Wed	8.0	Study ImgFi paper & image coding	Developed `make_images.py` for recurrence transformations	$RT[i,j] = \|x_i - x_j\|$, grayscale 224x224
July 9	Thu	8.0	Script logging & activity boundaries	Coded `session_logger.py` with discrete and continuous modes	Defined standard protocols for 7 target classes
July 10	Fri	6.0	Test verification gates	Ran MAC check and wiggle tests; resolved MAC drift	Gateway pinging stabilized MAC detection
July 13	Mon	6.5	Ambient capture & interval analysis	Captured `csi_noaction.csv`; checked packet spacing	Observed packet spacing jitter (up to 464 ms)
July 14	Tue	8.0	Code segmentation & resampling	Programmed `segment_windows.py` linear resampling	Standardized windows onto uniform 100 Hz grid
July 15	Wed	8.0	Collect multi-subject test data & QA	Captured test files; coded automated QA script `check_capture.py`	Automatically flags row drops, NaNs, and zero rows
July 16	Thu	8.0	Study Doppler shifts & sample rates	Checked Nyquist sufficiency; analyzed discriminative frequency bands	Doppler scaling: 16.4 Hz per m/s; signal band < 18 Hz
July 17	Fri	7.0	Collect official pilot dataset	Captured 8 official action/noaction CSVs (walk, stand, mixed, sit)	Checked all files passed QA checks

Date	Day	Hours	Planned Work Summary	Completed Work Summary	Key Parameters / Notes
July 20	Mon	8.5	Segment windows and train ML models	Segmented data at rate 50; trained ResNet-18 and CNN-BiGRU-Attention	CNN-BiGRU-Attention got 98.35% accuracy using 0.57 M parameters